

Syed Mustafa Hassan

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Education & Training

Bachelor of Science in Computer Science | Mohammad Ali Jinnah University (MAJU) | 11/10/2021 - 16/08/2025 | Karachi, Pakistan

• **Academic Honors:** Merit scholarship (2021-2025) and the Dean's List of Honor (2022-2024)

• **Highlighted Courses:** Generative AI (Large Language Model), Machine Learning, Computer Vision, Human-Computer Interaction, Data Structures & Algorithms, Design and Analysis of Algorithms, Applied Physics, Parallel Computing, Probability & Statistics, Linear Algebra, Numerical Computing

Final grade: 3.46/4.0 CGPA

Higher Secondary Certificate | Govt. Degree College Latifabad No.11 | 07/01/2019 - 22/07/2021 | Hyderabad, Pakistan

Final grade: A1 (91.81%)

Independent Study & Technical Development

Graduate Preparation | 01/09/2025 - 31/07/2026

• Pursued independent study in machine learning, computer vision, NeuroAI, and human visual processing to strengthen preparation for graduate study in data science and computational imaging.

• Strengthened practical skills in Python, PyTorch, TensorFlow, OpenCV, NumPy, Pandas, and SQL through structured coursework, experiments, and implementation-based learning.

• Expanded knowledge of computer networks, cloud infrastructure, Linux, and network fundamentals through Cisco and AWS coursework, developing a stronger understanding of connectivity, network services, and infrastructure management.

• Continued developing software engineering knowledge in databases, APIs, Git-based development, and deployment practices to better understand end-to-end data and software systems.

Work experience

Network Resident Engineer | Wateen Telecom Limited | 03/08/2026 - Current | Karachi, Pakistan

• Designed, implemented, and maintained LAN/WAN systems and network protocols to ensure smooth operation.

• Troubleshooted and resolved network issues promptly, minimizing downtime and optimizing network performance.

• Gain hands-on experience with enterprise network infrastructure and secure services, strengthening understanding of distributed systems, infrastructure reliability, and fault isolation.

Open Source Contributor | Social Winter of Code | 02/09/2024 - 29/11/2024

• Contributed to LLM-based applications by integrating APIs and databases to support real-time data access and application functionality.

• Worked with CI/CD pipelines and Git-based workflows to support collaborative development, testing, and code updates.

• Developed and integrated RESTful APIs to connect machine learning models with application components.

Front-end Developer Intern | 360XpertSolutions | 01/01/2024 - 30/03/2024 | Karachi, Pakistan

• Developed reusable and responsive UI components using React and Next.js, working with multiple libraries to build clean and user-friendly interfaces.

• Assisted with backend development using Nest.js, Node.js, and FastAPI, including REST and GraphQL API integration and implementation of application logic.

• Collaborated with the development team to troubleshoot issues, improve application performance, and deliver features across frontend and backend systems.

Selected Academic Projects

BrainCare – Dementia Patient Care Application (Final year project)

• Developed a React Native mobile application that supports caregivers of dementia patients.

• Caretakers can monitor track patient's location, manage medication, and access e-medical support.

• The patient panel includes an SOS feature, medication reminders, and brain exercise games.

VisionGlyco – Non-Invasive Glucose Monitoring System (Microsoft Imagine Cup)

• Built an end-to-end application for non-invasive glucose monitoring though capturing eye images using their smartphones.

• Implemented a real-time iris image inference pipeline using trained convolutional neural network model that process iris images within one second.

Low Light Image Enhancement (Generative AI coursework)

• Developed a lightweight UNet-based architecture with an attention mechanism that achieves 27.82 dB PSNR for real-time low-light image enhancement.

Quality Assessment of HDR Tone-Mapping Images (Human-computer interaction coursework)

• Objective quality assessment for HDR tone-mapped images using multi-scale structural similarity and natural image statistics.

• Evaluated eight different tone-mapping algorithms utilizing TMQI dataset.

3D Point Cloud Segmentation And Classification Using PointNet (Machine learning coursework)

• PointNet paper was re-implemented in PyTorch for the direct processing of 3D point clouds (using S3DIS dataset) that enables per-point semantic segmentation and classification without the need for intermediary representations.

Skills

Technical Expertise:

Languages: Python, MATLAB, C++, JavaScript, Java.

AI/ML: TensorFlow, PyTorch, OpenCV, scikit-learn, NumPy, Pandas.

Frontend: React, React Native, HTML5, CSS3.

Backend: Node.js, Express.js, RESTful APIs.

Databases: SQL, PostgreSQL, MongoDB, MySQL, Firebase.

Cloud & DevOps: AWS, Microsoft Azure, Git, GitHub, Linux.

Language Skills

Mother tongue(s): URDU

Other language(s):ENGLISH

Certification and Courses

The Human Brain - MIT OpenCourseWare (Ongoing)

Studied the fundamental principles of the human vision and cognition from retinal processing to high level visual streams. This includes about 17 hours of lectures.

NeuroAI course - Neuromatch Academy

Learned about the common principles between human brain and artificial intelligence, network architectures, and how AI models compare to biological neural systems with hands-on exercises.

Machine Learning Specialization - DeepLearning.AI & Standford Online

Gained necessary theoretical and mathematical foundations of machine learning involving supervised and unsupervised models. This course consisted of 94 hours workload.

Code in Place Program - Stanford University

Focused on programming fundaments in python with extensive practical training through implementing several projects.

SQL Fundamentals - DataCamp

Acquired core SQL concepts for data analysis, from basic querying to advanced techniques like window functions and database design. This course has a 26-hour commitment.

AWS Cloud Practitioner Essentials - Amazon Web Services

Developed knowledge of foundational AWS Cloud concepts with a focus on AWS Compute, Storage, Networking and Database services, security, architecture, pricing, and support.

Networking Basics - Cisco

Obtained knowledge of network types, data transmission, cabling, IP addressing, protocols, and built a wireless network via Cisco Packet Tracer.

NSE-1 Cybersecurity and Cloud Fundamentals - Fortinet

Learned essential digital threats, defense strategies, and the basics of securing cloud networks.

Honours & Awards

[2021-2025] Awarded merit based scholarship for BS (Computer Science) at Mohammad Ali Jinnah University

[2022-2024] Academic Performance Award Dean's List Roll Of Honor Spring

[2023-2025] Beta Microsoft Learn Student Ambassador

[2025] Selected for Stanford University Code in Place

[2024] Top Open Source Contributor in Google's Social Winter of Code

[2023] Third Position in Jinnah League Competitive Programming Competition

Extra Curricular Activities

Open Source Contributor:

GirlScript Summer of Code (2023), Script Winter of Code (2024), Hacktoberfest (2024)

Community Leadership:

Founder of Microsoft Learn Student Club at campus, Core Member – Google Developer Student Club (2022-2024), Microsoft Learn Student Ambassador (2022-2025), GitHub Global Campus Student (2023-2025), McKinsey Forward Program Apprentice (2024)

Competitive Programming:

ICPC (2024), Microsoft Imagine Cup (2024), Google Solution Challenge (2024), Meta Hacker Cup (2023)